

Output Devices — Detailed Lecture Notes

JKSSB Computer Awareness

October 8, 2026

14.1 What is an output device

A computer works in a cycle: input, process, and output. In the output step, the result of processing is handed back to the user or to another device. An **output device** is the hardware that receives processed data from the CPU and presents it in a form a human can see, hear, or read. Without an output device the result of the computation would stay locked inside the machine.

Definition (Output device). *Hardware that receives processed data from the CPU and presents it to the user or to another device in a readable form, such as on a screen, through a speaker, or on paper.*

Common output devices include the monitor, projector, speaker, headphones, printer, and plotter. The following table fixes the meaning of each term used in this lesson.

Term	Meaning in this lesson
Monitor	The main visual output device; shows text, images, and video as soft copy.
Projector	Throws a large image onto a screen or wall as soft copy.
Speaker	Converts an electrical signal into sound for a group of listeners.
Headphones	Produce sound for a single listener; worn on or over the ears.
Printer	Produces hard copy on paper; impact or non-impact.
Plotter	Draws large, precise line drawings such as engineering designs.
MFP	Multifunction Printer; combines printing, scanning, copying, and often faxing.

14.2 Classifying output devices

Output devices are grouped by the form in which they present the result. This is the classification JKSSB question-setters use most often, so learn one example for each row.

Category	Typical devices
Visual / soft copy	Monitor, projector
Audio	Speaker, headphones
Printed / hard copy	Printer, plotter

A single device may belong to more than one row if it produces more than one kind of output, but for exam purposes use the primary role: a monitor is a visual output device, a speaker is an audio output device, and a printer is a printed (hard copy) output device.

14.3 Soft copy vs hard copy

Output comes in two forms. **Soft copy** is output that is displayed electronically, such as an image on a monitor or a projected slide; it is not printed and can usually still be edited. **Hard copy** is output that is printed on a physical medium such as paper, produced by a printer or a plotter; once printed it is fixed.

Definition (Soft copy). *Output that is shown on a screen or projected electronically and is not printed, such as the image on a monitor or projector.*

Definition (Hard copy). *Output that is printed on a physical medium such as paper, produced by a printer or a plotter.*

The distinction matters because a single item can test it in either direction. If the stem mentions a screen, a display, or a projected image, the answer is soft copy; if it mentions paper, printing, or a physical printout, the answer is hard copy.

Point	Soft copy	Hard copy
Form	Electronic, on a screen	Physical, on paper
Examples	Monitor, projector	Printer, plotter
Editable	Easily edited	Fixed once printed

14.3.1 Uses and limits of each form

Soft copy and hard copy serve different needs, and an exam item may ask which is better for a stated purpose. Soft copy is convenient because it can be edited, searched, copied, and shared electronically, and it costs nothing to display; its limit is that it needs power and a screen, and it is easy to lose if it is not saved. Hard copy is convenient because it needs no power to read, is permanent, and can be signed or filed as a legal record; its limits are that it can be bulky, is fixed once printed, and consumes paper and ink. The rule of thumb is: to read, edit, or send, use soft copy; to sign, file, or archive, use hard copy.

14.4 Monitors

A **monitor** is the main visual output device of a computer. It displays text, images, and video as soft copy. Screen size is measured diagonally in inches, and picture sharpness is measured in pixels, which together give the resolution of the display. A monitor shows output to the user; it does not take input, with the single exception of a touchscreen, which senses touch in addition to displaying.

14.4.1 CRT displays

A **CRT** display uses a **Cathode Ray Tube**. An electron beam sweeps across a phosphor-coated screen inside a vacuum tube to form the image. CRT monitors are bulky and deep, and they are now largely replaced by flat panels. For exam purposes the key point is the full form: **CRT = Cathode Ray Tube**, and it is the older, bulky technology.

14.4.2 Flat-panel displays: LCD, LED, and OLED

A **flat-panel display** is thin, light, and flat. The common technologies are LCD, LED, and OLED.

Type	What it means	Backlight
LCD	Liquid Crystal Display; cells of liquid crystal control light	Needs a backlight
LED	Light Emitting Diode display; an LCD panel lit by LEDs	LED backlight
OLED	Organic Light Emitting Diode; each pixel emits its own light	No backlight (self-emitting)

The nearest-neighbour trap is LCD versus LED versus OLED. An **LCD** needs a separate backlight to be visible. An **LED** display is really an LCD panel whose backlight is made of LEDs. An **OLED** display needs no backlight at all, because each pixel makes its own light.

14.4.3 Display size, resolution, and refresh

The size of a monitor is given as the diagonal length of the screen in inches. The picture is made of tiny dots called **pixels**; the number of pixels across and down the screen is the **resolution**, such as 1920 by 1080. A higher resolution shows more detail in the same screen area. The **refresh rate** is how many times per second the screen is redrawn, measured in hertz (Hz); a higher refresh rate gives smoother motion. These are the three numbers a display specification quotes, and they apply to monitors and to laptop screens alike.

14.5 Projector

A **projector** throws a large image onto a screen or a wall. It is used in classrooms, meetings, and cinemas to show the same computer output to many people at once. The projected image is soft copy, because it is shown on a surface and not printed. A projector is an output device; it does not capture anything, so it must not be confused with a scanner or a camera.

14.6 Audio output: speakers and headphones

A **speaker** converts an electrical signal into sound that a group of listeners can hear. **Headphones** produce sound for a single listener and are worn on or over the ears. Both are audio output devices.

The nearest-neighbour pair is the speaker against the microphone. A **microphone** takes sound in and is an audio input device; a **speaker** or a **headphone** sends sound out and is an audio output device.

Example 14.6.1. *A student plays a recorded lecture on the office desktop. The sound leaves the computer through the speakers, so the speakers are audio output devices. If the student then records a reply into the same computer, the microphone is used; that is an audio input device. The same machine performs both jobs through two different devices.*

14.7 Printers: overview

A **printer** produces hard copy on paper. Printers fall into two broad classes according to how they mark the paper. An **impact printer** strikes a ribbon against the paper; a **non-impact printer** marks the paper without striking it, using ink, toner, or heat. The common types are the dot-matrix printer (impact) and the inkjet and laser printers (non-impact). Laser printers are common in offices, while inkjet printers are common in homes. A printer is an output device (TRAP-12).

Point	Impact	Non-impact
Method	Strikes a ribbon and the paper	No striking; uses ink, toner, or heat
Examples	Dot-matrix	Inkjet, laser
Noise	Noisier	Quieter
Graphics quality	Lower	Higher

A **dot-matrix** printer is an impact printer and is the noisiest of the common types. An **inkjet** printer sprays tiny drops of ink onto the paper, and a **laser** printer forms the image with toner that is fused onto the paper by heat. Both are non-impact printers.

The choice follows the job. A dot-matrix printer is cheap to run and can print multi-part carbon forms, but its quality is low and it is noisy. An inkjet printer is quiet and inexpensive to buy and is suited to colour photographs and home use, though the ink can be costly. A laser printer is fast, quiet, and gives sharp text, which is why it is the usual office choice. Whatever the type, every printer is an output device producing hard copy.

Example 14.7.1. *An office prints a hundred letters every morning. The best choice is a laser printer, because it is fast, quiet, and gives sharp text. If the same office had to print only the occasional colour photograph at home, an inkjet printer would be enough. If it needed multi-part carbon forms for a carbon copy, a dot-matrix printer would be the right impact printer.*

14.8 Plotter

A **plotter** draws large, precise line drawings, such as engineering and architectural designs, on paper larger than a normal printer's. It draws using a pen or a vector-based method, which makes it suitable for the continuous lines of technical drawings. A plotter is an output device: it produces hard copy but it does not read anything, so it must not be confused with a scanner (TRAP-12).

Example 14.8.1. *An item asks which output device is used to produce large engineering drawings. The answer is a **plotter**, because it is designed for large, precise line drawings. A normal printer is meant for ordinary pages, a monitor shows soft copy, and a scanner is an input device.*

14.9 Multifunction printer (MFP)

An **MFP** is a **Multifunction Printer**, also called an **all-in-one** device. It combines **printing**, **scanning**, **copying**, and often **faxing** in a single unit. Because it does several jobs, an MFP is the one common device that is both an input and an output device: its scanner part takes input, while its printer part gives output. MFPs are common in offices and small businesses.

14.10 Summary of output devices

The table below gathers the devices in this lesson for quick revision. Use it as a checklist: for each device, be ready to state its category (visual, audio, or printed) and its one-line function, and to name the device it is most often confused with.

Device	Category	Function
Monitor	Visual / soft copy	Shows text, images, and video on a screen.
Projector	Visual / soft copy	Throws a large image onto a screen or wall.
Speaker	Audio	Converts an electrical signal into sound for a group.
Headphones	Audio	Produces sound for a single listener.
Printer	Printed / hard copy	Prints text and graphics on paper; impact or non-impact.
Plotter	Printed / hard copy	Draws large, precise line drawings.
MFP	Hybrid	Combines printing, scanning, copying, and often faxing.

14.11 Points of confusion between similar devices

Beyond classifying a device, JKSSB items often test a single distinction between two devices that look or sound alike. The table below collects the pairs used in this lesson.

Pair	The distinction to state
Printer vs plotter	Printer: standard pages of text and graphics. Plotter: large, precise line drawings.
Monitor vs projector	Monitor: image on its own screen. Projector: image thrown onto a screen or wall.
Speaker vs microphone	Speaker: audio output. Microphone: audio input.
Soft copy vs hard copy	Soft copy: shown on a screen. Hard copy: printed on paper.
LCD vs OLED	LCD: needs a backlight. OLED: self-emitting, no backlight.
Impact vs non-impact	Impact: strikes the paper (dot-matrix). Non-impact: no striking (inkjet, laser).

Example 14.11.1. *An office buys one machine that must print letters, scan signed forms, and photocopy pages. The right choice is a Multifunction Printer (MFP), because it does all three jobs in one unit. If the office instead needed very large engineering plans, the correct device would be a plotter, and if it only needed to show a presentation to a room, the correct device would be a projector. Matching the output device to the job is the practical form of this topic's classification.*

14.12 Other output devices on the exam

A few output devices appear in real JKSSB items without being part of the main list. A **sound card** produces audio output: it turns the digital sound data into a signal for the speakers. A **Voice Response Unit (VRU)** speaks responses to a caller and so gives audio output. **Haptic gloves** and a **tactile display** give touch or force feedback to the user and are therefore output devices. All of these feed information *back* to the user and must be told apart from input devices of similar names, such as a **data glove**, which captures hand movement as input (PYQ-09).

Definition (Sound card). *An expansion card that produces audio output, converting digital sound data into a signal that speakers or headphones can play.*

14.13 Exam retrieval and common confusions

Seven distinctions prevent most errors on this topic.

1. **Soft copy** is on a screen (monitor, projector); **hard copy** is printed on paper (printer, plotter). Do not swap them.
2. A **plotter** and a **printer** are **output** devices; a **scanner**, a **light pen**, and a **digitizer** are **input** devices (TRAP-12).
3. **CRT** = Cathode Ray Tube and is bulky; modern monitors are flat panels (LCD, LED, OLED).
4. An **LCD** needs a backlight, an **LED** display is an LCD with an LED backlight, and an **OLED** display is self-emitting with no backlight.
5. A **dot-matrix** printer is an **impact** printer; inkjet and laser printers are **non-impact**.
6. A **plotter** draws large engineering drawings (PYQ-29), while a normal printer is meant for ordinary pages.
7. A **sound card**, a **Voice Response Unit (VRU)**, a **haptic glove**, and a **tactile display** give output to the user; a **data glove** captures hand movement as input (PYQ-09).

Example 14.13.1. *An item lists four devices — scanner, plotter, drum printer, and light pen — and asks which are output devices. Classify each one: the scanner is input, the plotter is output, the drum printer is output, and the light pen is input. The answer is the plotter and the drum printer. This is exactly the input/output classification tested by TRAP-12.*

Exercises

1. Define an output device and give three examples.
2. Distinguish soft copy from hard copy, giving one example of each.
3. Expand CRT, LCD, and OLED, and state which of them needs a backlight.
4. Explain the difference between an impact printer and a non-impact printer, giving one example of each.
5. Which output device is used for large engineering drawings, and why?
6. State whether each is input or output: plotter, scanner, projector, digitizer tablet, speaker.

Solutions

1. An output device receives processed data from the CPU and presents it to the user in a readable form. Examples: monitor, printer, speaker (any three valid devices).
2. Soft copy is shown on a screen and is not printed, such as an image on a monitor or projector. Hard copy is printed on paper, such as a page from a printer or plotter.
3. CRT = Cathode Ray Tube; LCD = Liquid Crystal Display; OLED = Organic Light Emitting Diode. The LCD needs a backlight; the OLED does not, because each pixel emits its own light.

4. An impact printer strikes a ribbon against the paper, such as a dot-matrix printer. A non-impact printer marks the paper without striking it, such as an inkjet or a laser printer.
5. A plotter is used, because it is designed to draw large, precise line drawings such as engineering and architectural designs (PYQ-29).
6. Plotter — output; scanner — input; projector — output; digitizer tablet — input; speaker — output.